Available online at www.sciencedirect.com

ScienceDirect

Energy Procedia 150 (2018) 74–82

Energy Procedia

www.elsevier.com/locate/procedia

12th International Photovoltaic Power Generation and Smart Energy Conference & Exhibition (SNEC 2018)

Competitiveness of PV Inverter as a Reactive Power Compensator considering Inverter Lifetime Reduction

Oktoviano Gandhi[a,b]*, Carlos Rodríguez-Gallegos[b,c], Thomas Reindl[b], Dipti Srinivasan[c]

[a]*Graduate School of Integrative Sciences and Engineering, National University of Singapore (NUS), Singapore 117456.*
[b]*Solar Energy Research Institute of Singapore (SERIS), National University of Singapore (NUS), Singapore 117574.*
[c]*Department of Electrical and Computer Engineering, National University of Singapore (NUS), Singapore 117576.*

Abstract

With the increasing adoption of photovoltaic systems (PVs) in distribution grid, many researchers and grid operators have proposed and started to utilise PV inverters for local reactive power compensation (RPC). The local RPC has been shown to reduce losses in the system, and to help maintain voltage within acceptable range. Moreover, RPC using PV is already competitive when compared with traditional reactive power devices. Nevertheless, the RPC using a PV inverter increases the current flowing through it, and hence the losses and the temperature of its components. As a result, the lifetime of the inverter will be reduced with increasing reactive power usage, incurring costs to the system owner and increases the PV levelised cost of electricity (LCOE). It is therefore imperative to consider the inverter degradation, as a rise in the LCOE may cause PV to be less competitive compared to other energy sources. Thus, in this work, the competitiveness of PV inverter as a reactive power compensator is reassessed, accounting for the inverter lifetime reduction. Case studies on test systems based on real distribution network are conducted to illustrate the importance of considering inverter lifetime reduction. The results from this work can be used by PV owners, power system operators and regulators in determining whether to utilise PV for RPC, and to what extent, based on their market situation.

Keywords: distribution system; inverter lifetime reduction; multiobjective optimisation; photovoltaic system; reactive power; switched capacitor

* Corresponding author. *E-mail address:* oktoviano.gandhi@u.nus.edu

10.1016/j.egypro.2018.09.005

1. Introduction

In recent years, solar energy, especially in the form of photovoltaic systems (PVs), has been increasingly used to provide electricity. As PV produces electricity at direct current (DC), it needs to be interfaced with an inverter that converts its electricity into alternating current (AC) present in most electrical grids. These inverters make it possible for PV to not only produce active power, but also reactive power. Many researchers have proposed the use of reactive power to minimize losses in the system [1], [2] and to maintain voltage within acceptable levels [3], [4]. Moreover, line loading and system operational costs can also be reduced through the local reactive power provision using PV [5].

Recognizing the potential benefits brought about by PV and other inverter-based energy sources as distributed reactive power sources, California's Electric Tariff Rule 21 and IEEE Standard 1547a have allowed smart inverters to provide reactive power compensation [6], [7] beyond their previously accepted contributions.

Nevertheless, few works have analysed the cost of reactive power from PV inverters, or the necessary incentives required for the PV owners. The work in [8] have proposed reactive power market framework for conventional generators. Subsequently, [2] have recognised the need to remunerate the reactive power provision from PV and other distributed energy resources. Nevertheless, the aforementioned references did not analyse what the costs are, and the economic viability of reactive power compensation (RPC) from inverters remains unclear. However, such analysis is of utmost importance to determine whether power system operators should utilise PVs as reactive power compensators, and whether the PV owners should inject/absorb reactive power given the incentives.

The authors' recent work [9] has analysed the competitiveness of PV inverters for RPC in distribution system and compared them to switched capacitors (SC). It has been shown that PV becomes increasingly competitive with SC at higher penetration levels and higher inverter efficiency. Nevertheless, [9] has not taken into account the inverter lifetime reduction due to reactive power usage recently formulated in [10].

Therefore, this paper reassesses the competitiveness of PV inverters as reactive power compensators by considering the inverter lifetime reduction due to RPC. Multi-objective optimisation has been employed to analyse the technical and economic benefits of the different reactive power compensators. In this work, the importance of considering inverter lifetime reduction in reactive power provision and optimization is demonstrated. Additionally, the work explores the effectiveness of combining both PV and SC for local RPC.

The rest of the paper is as follows: Section 2 describes the problem, optimisation objectives and constraints. Section 3 elaborates the parameters used in the case studies, followed by the information on the algorithm employed in Section 4. The results and discussions are presented in Section 5 and finally Section 6 concludes the work.

2. Problem formulation

2.1. Objective function

Economic reactive power dispatch is generally concerned with minimizing the operational cost of running the electrical grid while fulfilling the power system constraints [11], [12]. At the same time, reactive power compensation has been frequently used to minimize the voltage deviation [13], [14]. Therefore, the objective functions adopted in this work are the maximisation of the net monetary benefits (NMB) from reactive power compensation (economic objective), and of the voltage improvement (technical objective).

Before we can define the NMB of RPC, the total costs of operating the system needs to be elaborated. The total cost of running a grid-connected distribution system with PVs can be expressed as:

$$\text{Cost} = \sum_{t=1}^{T} \left[\underbrace{c_t^{\text{Pgrid}} P_t^{\text{grid}}}_{\text{cost of P from grid}} + \underbrace{c^{\text{Qgrid}} Q_t^{\text{grid}}}_{\text{cost of Q from grid}} + \underbrace{\sum_{x=1}^{M} c_x^{\text{PPV}} P_{x,t}^{\text{PV}}}_{\text{cost of P from PV}} + \underbrace{\sum_{x=1}^{M} c_{x,t}^{\text{QX}} Q_{x,t}^{\text{X}}}_{\text{cost of Q from PV/SC}} \right] H \tag{1}$$

where P_t^{grid} (Q_t^{grid}) is the active (reactive) power taken from the grid, while $P_{x,t}^{\text{PV}}$ is the active power generated by PV in the distribution system. $Q_{x,t}^{\text{X}}$ refers to either reactive power output from the PV inverters or from SC, depending on the case being analysed. c_t^{Pgrid} is the electricity price purchased from the grid, and c^{Qgrid} is the grid reactive power charge. The payment to the PV owners for their active power contribution is represented by c_x^{PPV}. $c_{x,t}^{\text{QX}}$ can be replaced by $c_{x,t}^{\text{QPV}}$ or $c_{x,t}^{\text{QSC}}$ to indicate the reactive power cost from PV or SC, respectively. The index t indicates quantities at period t from 1 to T. H [hour] is the length of the time period considered in the formulation. The subscript x and superscript X refer to quantities related to either PV or SC.

The NMB from RPC using PV is thus the difference in Cost when there is no RPC from PV and when there is:

$$\text{NMB} = \text{Cost}^{\text{NoQ}} - \text{Cost}^{\text{X}} \tag{2}$$

where the NoQ superscript refers to the case where there is no RPC in the system, while the X superscript refers to the case with RPC either from PV or SC, as explained in more detail in Section 3.1.

On top of the economic objective, it is also important to assess the RPC in terms of the reduction in voltage deviation, which can be measured through a quantity called weighted voltage deviation index (WVDI), defined as:

$$\text{WVDI} = \sum_{t=1}^{T}\sum_{i=1}^{N}\left(V_{i,t} - V_{\text{ref}}\right)^2 \tag{3}$$

where $V_{i,t}$ is the voltage at node i, and V_{ref} is the reference voltage of the system. N is the number of nodes in the system. Therefore, the technical objective of the RPC can be expressed through the following:

$$\text{WVDI Improvement} = \text{WVDI}^{\text{NoQ}} - \text{WVDI}^{\text{X}} \tag{4}$$

The power dispatch optimization is subject to the general power system constraints [3].

2.2. PV model

The power generated from a PV system, P_t^{PV} – given global irradiance G_t , cell temperature T_t^{PV}, and the PV rating at standard test conditions $P_{\text{rated}}^{\text{PV}}$ – is expressed by the following equation [15]:

$$P_t^{\text{PV}} = \frac{G_t P_{\text{rated}}^{\text{PV}}\left(1-\eta^{\text{T}}\left(T_t^{\text{PV}} - 25\right)\right)}{1000} - P_t^{\text{PV,invloss}} \tag{5}$$

where $P_t^{\text{PV,invloss}}$ is the inverter active power loss in the inverter. η^{T} and NOCT are the power temperature coefficient and the nominal operating temperature of the solar cell respectively.

Given P_t^{PV}, the PV system can generate reactive power according to the following constraint:

$$\left|Q_{x,t}^{\text{PV}}\right| \le \sqrt{\left(S_{\text{max}}^{\text{PV}}\right)^2 - \left(P_{x,t}^{\text{PV}} + \Delta P_{x,t}^{\text{PV}}\right)^2} \tag{6}$$

where $S_{\text{max}}^{\text{PV}}$ is the PV inverter rating, while $\Delta P_{x,t}^{\text{PV}}$ is the uncertainty in PV power forecast. Constraint (6) is imposed so as not to reduce the active power generated by the PV, which generally yields higher monetary value.

Cost of reactive power is divided into two components: the inverter loss and the inverter degradation/lifetime reduction [10]:

$$c_{x,t}^{\text{QPV}} = c_{x,t}^{\text{QPV,invloss}} + c_{x,t}^{\text{QPV,LR}} \tag{7}$$

where $c_{x,t}^{\text{QPV,invloss}}$ and $c_{x,t}^{\text{QPV,LR}}$ are the unit cost of the reactive power due to inverter loss and inverter lifetime reduction (LR), respectively. The former arises from the additional power losses in the inverter due to the increase in the current flowing through it. The additional power losses have to be compensated and have the following form [3]:

$$c_{x,t}^{\text{QPV,invloss}} Q_{x,t}^{\text{PV}} = c_t^{\text{Pgrid}} \times \Delta P_{x,t}^{\text{PV,invloss}} \tag{8}$$

where $\Delta P_t^{\mathrm{PV,invloss}}$ is the additional losses in the inverter because of reactive power injection. The equations to calculate $\Delta P_t^{\mathrm{PV,invloss}}$ can be found in [3].

The second term in the right-hand side of Eq. (7) arises due to the LR of the inverter because of the reactive power usage. The increase in current flowing through the inverter induces additional losses, which in turn cause the temperature of the inverter components to rise [16]. Electrolytic DC-link capacitors are found to be most affected by the rise in temperature and are likely to fail sooner than other components [17], [18]. Therefore, it has been assumed that the lifetime of the inverter is the same as the lifetime of the DC-link capacitors.

When the inverter fails within the lifetime of the system (taken to be 20 years in this work), it has to be replaced. A shorter inverter lifetime due to RPC means that the inverter has to be replaced sooner and more frequently, hence increasing the levelised cost of electricity (LCOE) of the PV system. This increase in LCOE can therefore be translated to be the inverter LR component reactive power cost from PV, $c_{x,t}^{\mathrm{QPV,LR}}$, and needs to be compensated appropriately. Although a year-long simulation is necessary to estimate the ILR cost, the work in [10] has correlated $c_{x,t}^{\mathrm{QPV,LR}}$ and $Q_{x,t}^{\mathrm{PV}}$ for every half-hour period. This yields a cubic function of the following form [10]:

$$c_{x,t}^{\mathrm{QPV,LR}} Q_{x,t}^{\mathrm{PV}} = \eta_x^{\mathrm{LR},0} + \eta_x^{\mathrm{LR},1} Q_{x,t}^{\mathrm{PV}} + \eta_x^{\mathrm{LR},2} \left(Q_{x,t}^{\mathrm{PV}}\right)^2 + \eta_x^{\mathrm{LR},3} \left(Q_{x,t}^{\mathrm{PV}}\right)^3 \tag{9}$$

where $\eta_x^{\mathrm{LR},0}$, $\eta_x^{\mathrm{LR},1}$, $\eta_x^{\mathrm{LR},2}$, and $\eta_x^{\mathrm{LR},3}$ are the coefficients of the function that fit the cost of ILR due to the reactive power usage. The coefficients depend on the type of the inverter, as well as on the PV LCOE.

2.3. Switched capacitors' model

The switched capacitors (SCs) in this work are assumed to consist of three capacitor banks of the same capacity, such that there are four different operating points for each SC. Unlike PV, the main function of SC is to provide RPC to the system [19]. As such, $Q_{x,t}^{\mathrm{SC}}$ is not limited by Eq. (6).

3. Simulation setup

3.1. Test cases

To assess the competitiveness of PV as a reactive power compensator, three reference cases are set, namely:

NoQ Case: The system has PVs installed, which are generating active power. However, there is no local RPC in the system. The operational cost and WVDI of NoQ case are used in Eq. (2) and (3) to assess the NMB and WVDI improvement of all the cases described below.

PV Base Case: In addition to generating active power, the PVs are also generating reactive power, whose values are optimized by the optimisation method explained in Section 4.

SC Base Case: The PVs are not allowed to provide any RPC. In their place, SCs of the same capacity are optimized to do so. The SCs are assumed to be in the same location as the PVs to allow direct comparison.

Based on the parameters in the base cases, sensitivity analyses are conducted. From [9], two of the most important factors affecting the competitiveness of PV and SC for RPC are the electricity price and the inverter efficiency (only on PV case). Therefore, the impact of the two variables are analysed in detail in this work. A case with higher PV penetration is also considered to simulate the scenario when PVs can fulfil as much reactive load in the system as SCs (in the base case, reactive load fulfilment by SC is much higher than PV as $Q_{x,t}^{\mathrm{PV}}$ is limited by Eq. (6)). For each change in variable, the comparison is made with the NoQ case with the respective change in variable.

On top of the sensitivity analyses, this work also highlights the importance of considering the ILR cost. This is done by optimizing the PV base case twice, with different reactive power cost considerations:

1. PV Base Case: Cost in Eq. (1) considers both $c_{x,t}^{\mathrm{QPV,invloss}}$ and $c_{x,t}^{\mathrm{QPV,LR}}$
2. PV Base w/o ILR Case: Cost in Eq. (1) only considers $c_{x,t}^{\mathrm{QPV,invloss}}$

The ILR cost is also incorporated to the results found in the PV Base w/o ILR to illustrate its importance.

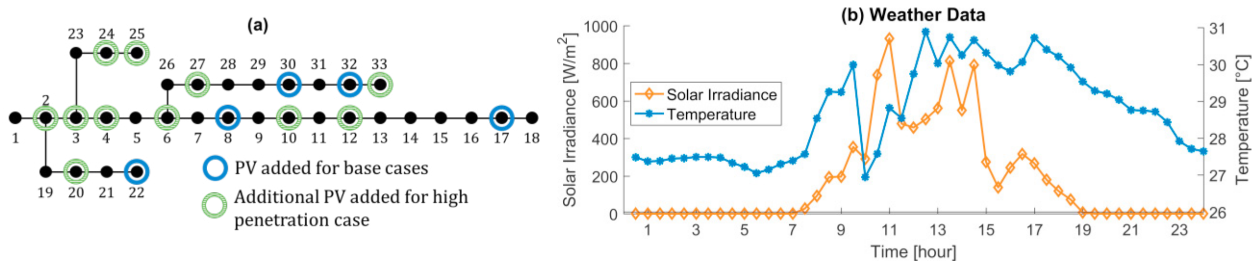

Fig. 1. (a) A 33-bus radial distribution system with PV added at the identified nodes, (b) Weather data used in the case studies.

A 33-bus test system [20] has been employed to analyse the competitiveness of PV inverters as reactive power compensators. As can be seen in Fig. 1 (a), the PVs are located at node 8, 17, 22, 30, and 32 for the base cases. When considering higher penetration of PV, there are 16 PVs in total, whose locations are also shown in Fig. 1 (a).

3.2. Power system and weather parameters

LCOE of PV, c^{PPV}, has been calculated based on the formula and parameters from [21] for the case of Singapore. The c^{QSC} is calculated based on the installation and maintenance cost from [22], and financial parameters, such as debt ratio and discount rate, from [21], resulting in value of 0.00248 SGD/kvarh. The load profile is taken from Singapore load data on 28th September 2017 [23]. c_t^{Pgrid} is the uniform Singapore electricity price, also on 28th September 2017, plus the associated grid charges [23], whereas c^{Qgrid} is the reactive power charge for customers taking power at 22 kV or 6.6 kV in Singapore [24].

$S_{\max}^{\mathrm{PV}}$ takes the value of 300 kVA. The values of $\eta_x^{\mathrm{LR},0}$, $\eta_x^{\mathrm{LR},1}$, $\eta_x^{\mathrm{LR},2}$, and $\eta_x^{\mathrm{LR},3}$ are 0, 0.213, -1.986, and 27.263 respectively [10]. V_{ref} is set at 1 p.u. The length of each time slot, H, is 0.5 hour, which makes T to be 48.

The weather parameters were also measured on 28th September 2017, at latitude 1.3002° and longitude 103.7715°. They are measured at every second by Solar Energy Research Institute of Singapore (SERIS) and are averaged into half-hour data to suit the price and load information data. The irradiance and temperature data are shown in Fig. 1 (b).

4. Algorithm

There are two algorithms used in this work: genetic algorithm (GA) for single-objective optimization and non-dominated sorting genetic algorithm-II (NSGA-II) [25] for multi-objective optimization. These two algorithms have been chosen because of their wide usage in the field [19], [26]–[28], and their capability in handling both discrete and continuous variables [29]. For detailed descriptions on the algorithms, the reader is referred to [25], [29].

NSGA-II is employed to analyse the competitiveness of PV in providing RPC compared to SC and to conduct the sensitivity analyses. Meanwhile, GA is used to maximise the NMB of PV Base Case and PV Base w/o ILR Case to analyse the importance of the ILR cost consideration. Both algorithms have 120 chromosomes and 60 generations

5. Results and Discussion

5.1. Impact of considering inverter lifetime reduction

First, the base case for PV and SC (using the parameters in Section 3) are presented in Fig. 2. Three optimisation results are presented, namely SC Base Case, PV Base Case, PV Base w/o ILR Case. The ILR cost were also incorporated to the PV Base w/o ILR results, which are then labelled as "PV Base w/o ILR + ILR" in Fig. 2.

From Fig. 2, it can be seen that for the base case, SC is more competitive for RPC compared to PV, regardless of the ILR cost consideration. This is mainly because SC is able to fulfil more reactive load, hence reduces more losses in the system, and push the voltage profile closer to V_{ref}.

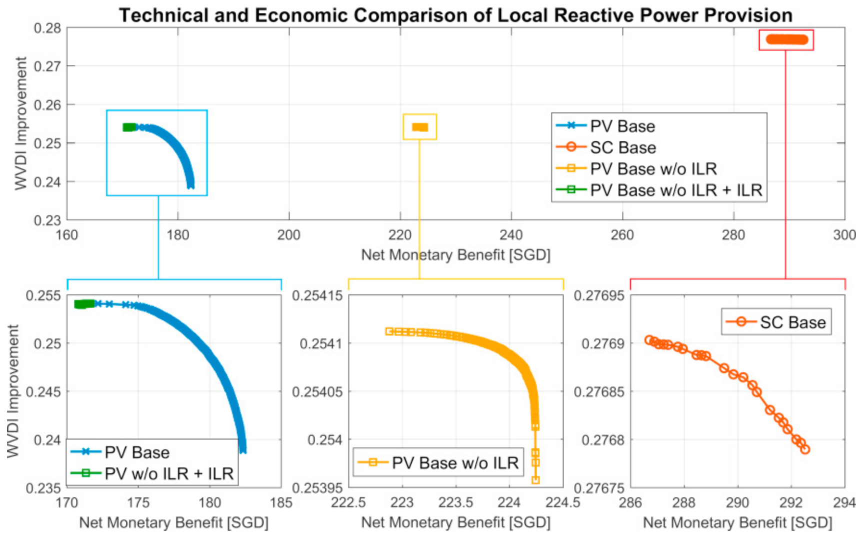

Fig. 2. Technical and economic comparison of local reactive power provision. The three panels below the main panel illustrate the shape of the pareto front of each of the case.

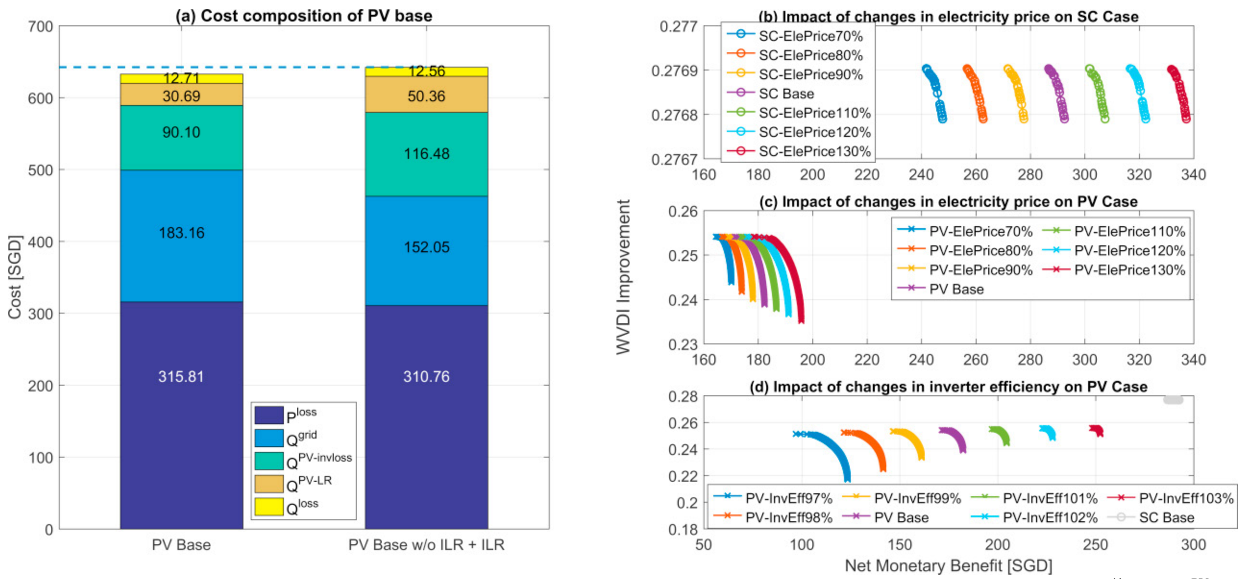

Fig. 3. (a) Cost composition of the most economical solution in PV Base (left) and PV Base w/o ILR + ILR (right). The cost of P^{grid} and P^{PV} are not shown because they are the same for the two cases. Both P^{loss} and Q^{loss} are fulfilled by the grid. They are separated to highlight the power losses in the system. (b) Impact of changes in electricity price on SC cases, and on (c) PV cases. (d). Impact of changes in inverter efficiency on PV cases. The InvEff 97%, 98%, 99%, Base, 101%, 102%, and 103% refer to peak inverter efficiency of 93.4%, 94.3%, 95.3%, 96.3%, 97.2%, 98.2% and 99.1%, respectively.

When the effect of reactive power usage on the inverter lifetime is not considered, the benefits of the RPC are overestimated by over 25%. By including the LR cost in Eq. (1), the multi-objective optimization yields larger pareto front, as there are more tradeoffs between the technical and economic objective when $c_{x,t}^{\text{QPV}}$ is higher.

To analyse the significance of considering the ILR cost, cost composition of the results that maximise the NMB from PV Base and PV Base w/o ILR + ILR is presented in Fig. 3 (a). It is shown that the result from PV Base w/o ILR reduces the power losses and Q^{grid} as the PV provides higher RPC compared to PV Base. Nevertheless, this brings about higher cost of reactive power, for both $c_{x,t}^{\text{QPV,invloss}}$ and $c_{x,t}^{\text{QPV,LR}}$. Because of that, the overall cost of PV Base w/o ILR is higher than that of the PV Base.

When ILR is not considered, $c_{x,t}^{\mathrm{QPV,LR}}$ is approximately 43% of $c_{x,t}^{\mathrm{QPV,invloss}}$. Thus, neglecting ILR cost in calculating the remuneration to the PV owners for local RPC will cause the PV owners to be underpaid.

5.2. Effect of change in electricity price and inverter efficiency

Each percentage change in electricity price causes approximately 0.5% and 0.24% change in NMB for the SC and PV cases respectively (Fig. 3 (b) and (c)) This occurs because at higher electricity prices, the loss reduction due to RPC becomes more valuable. At higher electricity price, reactive power cost from PV also increases. As such, there is more tradeoff in generating reactive power using PV, resulting in larger pareto front.

Inverter efficiency increases the competitiveness of RPC from PV significantly – a 14% change in NMB for every 1% increase in inverter efficiency – as shown in Fig. 3 (d). Nevertheless, even when the peak inverter efficiency is 99.1%, the RPC from SC is still more competitive.

5.3. Conditions required for PV to be competitive

In the previous scenarios, SC cases have higher technical and economic benefits as the SCs are able to fulfill more reactive load compared to PV. Therefore, a case with higher penetration of PV is also analysed. The effects of higher PV penetration on the competitiveness of PV for RPC are twofold. Firstly, higher penetration of PV increases its reactive power capability. Consequently, the PVs have more operating points to reduce the line losses and maintain the voltage near the reference point. Secondly, higher PV penetration also reduces the LR of the inverter, as the reactive power fulfillment is distributed over more PV and each PV generates lower reactive power [10].

At higher PV penetration and peak inverter efficiency of 98.2%, PV is already competitive in terms of the economic objective as seen in Fig. 4 (a). Although it appears that PV still lacks in the technical objective (Fig. 4 (a)), the WVDI improvement of PV is only worse than that of SC at 11:00, 13:30, and 14:30 (Fig. 4 (b)). The three periods coincide with the periods when the irradiance is high and PV is generating high active power (Fig. 1 (b)), such that the reactive power capability of the PV is lower. For the rest of the day, it can be seen that PV is always performing better than SC. The general better performance of PV comes from the continuous values of reactive power provided by the PV inverters, as opposed to the discrete values of SC.

It is also worth exploring to use a combination of PV and SC for RPC, i.e. SC can be used to fulfill the base reactive load while PV is used to fine tune the RPC and response to rapid variation in voltage and/or load. Fig. 5 shows the results of the RPC from combination of sources, for both the Base case as well as for the high penetration case. When RPC comes from a mix of PV and SC, the pareto fronts become larger than that of SC. The WVDI improvement of the 1PV-4SC case is higher than both SC and PV base cases, whereas its NMB is higher than that of PV case.

From Fig. 5 (b), it can be seen that replacing some of the SCs with PVs as the reactive power source enhances the NMB of the system without sacrificing the WVDI improvement. These results show that a combination of both PV and SC for RPC can be better than cases with only PV or SC.

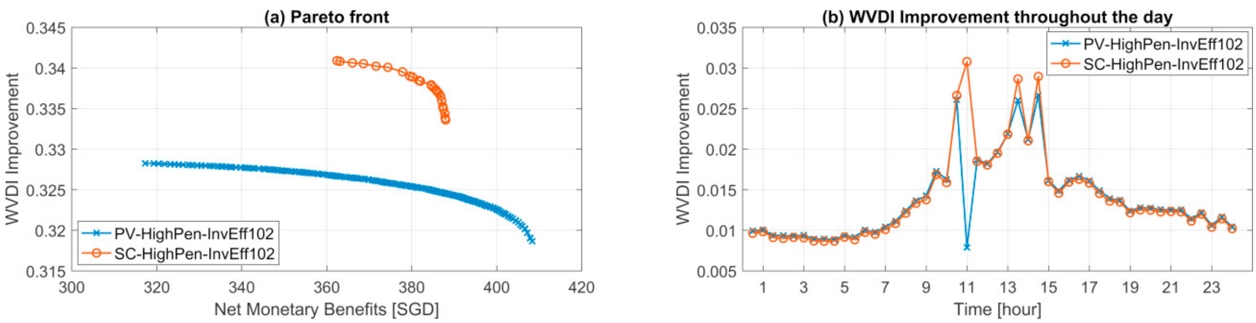

Fig. 4 Results of PV and SC cases considering high PV penetration and inverter efficiency 102% of the base value (peak inverter efficiency of 98.2%).

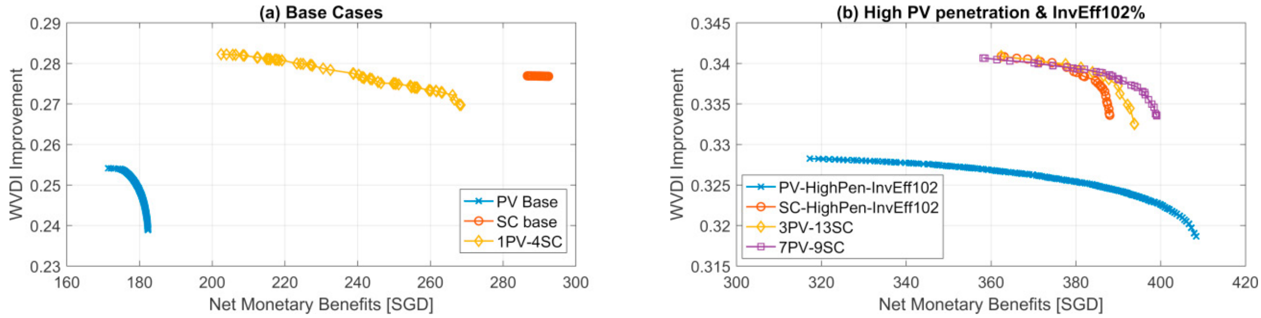

Fig. 5. Pareto fronts comparison between the results of PV and SC cases with the results from combinations of PV and SC, (a) for the base case, and (b) for the higher PV penetration and higher inverter efficiency. The notation 'X'PV-'Y'SC indicates that the RPC comes from X number of PV and Y number of SC.

6. Conclusion

With increasing photovoltaic system (PV) penetration across the world, assessment of its capability to provide ancillary services for the power system has become a hot topic. Local reactive power compensation (RPC) is one of the prominent ways in which PV can benefit a distribution system through the reduction in line losses and improved voltage control. Building up on previous works which formulated the cost of reactive power from PV inverters, this work has highlighted the importance of considering the inverter degradation due to RPC. Subsequently, the competitiveness of PV as reactive power compensator compared to switched capacitors (SC) has been reassessed. While PV is not as competitive as SC in providing RPC at low PV penetration, it becomes increasingly so at higher penetration. A combination of PV and SC can also be used effectively to provide local RPC for the system.

Acknowledgements

The Solar Energy Research Institute of Singapore (SERIS) is a research institute at the National University of Singapore (NUS). SERIS is supported by the National University of Singapore (NUS), National Research Foundation Singapore (NRF), and the Singapore Economic Development Board (EDB).

References

[1] O. Gandhi, W. Zhang, C. D. Rodríguez-Gallegos, D. Srinivasan, and T. Reindl, "Continuous optimization of reactive power from PV and EV in distribution system," in *2016 IEEE Innovative Smart Grid Technologies - Asia (ISGT-Asia)*, 2016, pp. 281–287.
[2] V. Kekatos, G. Wang, A. J. Conejo, and G. B. Giannakis, "Stochastic Reactive Power Management in Microgrids With Renewables," *IEEE Trans. Power Syst.*, vol. PP, no. 99, pp. 1–10, 2014.
[3] O. Gandhi, C. D. Rodríguez-Gallegos, W. Zhang, D. Srinivasan, and T. Reindl, "Economic and technical analysis of reactive power provision from distributed energy resources in microgrids," *Appl. Energy*, vol. 210, pp. 827–841, Jan. 2018.
[4] S. Deshmukh, B. Natarajan, and A. Pahwa, "Voltage/VAR control in distribution networks via reactive power injection through distributed generators," *IEEE Trans. Smart Grid*, vol. 3, no. 3, pp. 1226–1234, 2012.
[5] T. Sousa, H. Morais, Z. Vale, and R. Castro, "A multi-objective optimization of the active and reactive resource scheduling at a distribution level in a smart grid context," *Energy*, vol. 85, pp. 236–250, 2015.
[6] IEEE standards Coordinating Committee 21, "1547 IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems Amendment 1," 2014.
[7] Southern California Edison, "Electric Rule 21." 2017.
[8] K. Bhattacharya and J. Zhong, "Reactive Power as an Ancillary Service," *IEEE Trans. Power Syst.*, vol. 16, no. 2, pp. 294–300, 2001.
[9] O. Gandhi, D. Srinivasan, C. D. Rodríguez-Gallegos, and T. Reindl, "Competitiveness of reactive power compensation using PV inverter in distribution system," in *2017 IEEE PES Innovative Smart Grid Technologies Conference Europe (ISGT-Europe)*, 2017, pp. 1–6.
[10] O. Gandhi, C. D. Rodríguez-Gallegos, N. B. Y. Gorla, M. Bieri, T. Reindl, and D. Srinivasan, "Reactive Power Cost from PV Inverters Considering Inverter Lifetime Assessment," *IEEE Trans. Sustain. Energy*, pp. 1–1, 2018.
[11] A. Gabash and P. Li, "Active-reactive optimal power flow in distribution networks with embedded generation and battery storage," *IEEE Trans. Power Syst.*, vol. 27, no. 4, pp. 2026–2035, 2012.
[12] J.-Y. Park, J.-M. Sohn, and J.-K. Park, "Optimal Capacitor Allocation in a Distribution System Considering Operation Costs," *IEEE Trans.*

Power Syst., vol. 24, no. 1, pp. 462–468, 2009.
[13] A. Kargarian, M. Raoofat, and M. Mohammadi, "Reactive power market management considering voltage control area reserve and system security," *Appl. Energy*, vol. 88, no. 11, pp. 3832–3840, 2011.
[14] A. Savić and Ž. Durišić, "Optimal sizing and location of SVC devices for improvement of voltage profile in distribution network with dispersed photovoltaic and wind power plants," *Appl. Energy*, vol. 134, pp. 114–124, 2014.
[15] C. D. Rodríguez-Gallegos, O. Gandhi, D. Yang, M. S. Alvarez-Alvarado, W. Zhang, T. Reindl, and S. K. Panda, "A Siting and Sizing Optimization Approach for PV–Battery–Diesel Hybrid Systems," *IEEE Trans. Ind. Appl.*, vol. 54, no. 3, pp. 2637–2645, May 2018.
[16] A. Anurag, Y. Yang, and F. Blaabjerg, "Thermal Performance and Reliability Analysis of Single-Phase PV Inverters with Reactive Power Injection Outside Feed-In Operating Hours," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 3, no. 4, pp. 870–880, 2015.
[17] H. Wang, M. Liserre, F. Blaabjerg, P. De Place Rimmen, J. B. Jacobsen, T. Kvisgaard, and J. Landkildehus, "Transitioning to physics-of-failure as a reliability driver in power electronics," *IEEE J. Emerg. Sel. Top. Power Electron.*, vol. 2, no. 1, pp. 97–114, 2014.
[18] D. Ton and W. Bower, "Summary report on the DOE high-tech inverter workshop," 2005.
[19] C. H. Antunes, D. F. Pires, C. Barrico, Á. Gomes, and A. G. Martins, "A multi-objective evolutionary algorithm for reactive power compensation in distribution networks," *Appl. Energy*, vol. 86, no. 7–8, pp. 977–984, 2009.
[20] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *Power Deliv. IEEE Trans.*, vol. 4, no. 2, pp. 1401–1407, 1989.
[21] M. Bieri, K. Winter, S. Tay, A. Chua, and T. Reindl, "An irradiance-neutral view on the competitiveness of life-cycle cost of PV rooftop systems across cities," *Energy Procedia*, 2017.
[22] S. X. Chen, Y. S. F. Eddy, H. B. Gooi, M. Q. Wang, and S. F. Lu, "A centralized reactive power compensation system for LV distribution networks," *IEEE Trans. Power Syst.*, vol. 30, no. 1, pp. 274–284, 2015.
[23] EMC, "Energy Market Price Information." [Online]. Available: https://www.emcsg.com/marketdata/priceinformation. [Accessed: 01-Apr-2018].
[24] EMC, "Use of System Charges," 2016. [Online]. Available: https://www.mypower.com.sg/documents/ts-usc.pdf. [Accessed: 01-Apr-2018].
[25] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, 2002.
[26] O. Gandhi, C. D. Rodríguez-Gallegos, and D. Srinivasan, "Review of optimization of power dispatch in renewable energy system," in *2016 IEEE Innovative Smart Grid Technologies - Asia (ISGT-Asia)*, 2016, pp. 250–257.
[27] O. Gandhi, W. Zhang, C. D. Rodríguez-Gallegos, M. Bieri, T. Reindl, and D. Srinivasan, "Analytical Approach to Reactive Power Dispatch and Energy Arbitrage in Distribution Systems with DERs," *IEEE Trans. Power Syst.*, pp. 1–1, 2018.
[28] C. D. Rodríguez-Gallegos, J. P. Singh, J. M. Y. Ali, O. Gandhi, S. Nalluri, A. Kumar, V. Shanmugam, M. L. Aguilar, M. Bieri, T. Reindl, and S. K. Panda, "PV-GO : A multiobjective and robust optimization approach for the grid metallization design of Si-based solar cells and modules," *Prog. Photovoltaics Res. Appl.*, pp. 1–23, 2018.
[29] R. Chelouah and P. Siarry, "A Continuous Genetic Algorithm Designed for the Global Optimization of Multimodal Functions," *J. Heuristics*, vol. 6, pp. 191–213, 2000.